

$$A = 0.71 \text{ mm}$$

- 17 C Unpolarised light after passing through a polarizer, its intensity is halved (therefore we eliminate options A/B).

Amplitude remains at A. (The component of E field perpendicular to the polariser axis is absorbed (e.g. A_y is absorbed), leaving the transmitted light having amplitude A_x)

Since the polarization angle is 75° (or 105°), by resolving the electric field,

$$A' = A \cos 75^\circ = A \sin 15^\circ$$

19. **A** First minima position: